Chapter 11

Systems of Conservation Laws

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 11; printed p. 571; PDF page 588.

Chinese reference: 中译本对应章：守恒律组，书内第 404 页起。

Editorial reconstruction. This chapter develops wave geometry, the local Riemann solver, Riemann invariants, and entropy selection. Exposition, expanded proofs, Chinese annotations, problem solutions, and diagrams are reconstructed study material. Source theorem numbers are stated separately. External input and the scope of each existence assertion are specified explicitly.

11.1 Introduction

11.1.1 Conserved variables and the weak formulation

Let $\Omega \subset \mathbb{R}^m$ be an open state domain and let $F \in C^3(\Omega; \mathbb{R}^m)$. We study

$$u_t + F(u)_x = 0, \qquad u(x,0) = g(x), \quad x \in \mathbb{R}. \tag{11.1}$$

For a smooth solution, integration over $[a,b]$ gives $\frac{d}{dt}\int_a^b u\,dx = F(u(a,t)) - F(u(b,t))$. This is a vector balance: each component is a conserved density. In several space dimensions the flux is an $m \times n$ matrix and its divergence is taken row by row.

Definition 11.1 (Integral solution; source §11.1.1)

A bounded measurable u with essential range in a compact subset of Ω is an integral solution if, for every $\varphi \in C_c^\infty(\mathbb{R} \times [0,\infty); \mathbb{R}^m)$,

$$\int_0^\infty \int_{\mathbb{R}} [u \cdot \varphi_t + F(u) \cdot \varphi_x]\,dx\,dt + \int_{\mathbb{R}} g \cdot \varphi(x,0)\,dx = 0. \tag{11.2}$$

A strong L^1_{loc} initial trace, when required, is an additional property beyond this distributional initial condition.

守恒变量的选择

浅水取 $u = (h, q)$、$q = hv$。速度 v 是原始变量而不是动量密度。光滑时可作变量替换，跨激波时必须回到原守恒律的跳跃条件。

Theorem 11.2 (Jump condition; source §11.1.1, equation (20))
Suppose u is piecewise C^1 across $x = s(t)$, $s \in C^1$, with one-sided traces, and solves the PDE on each side. Put $[u] = u_r - u_l$, $[F] = F(u_r) - F(u_l)$. The interior weak equation holds if and only if

$$[F] = s'(t)[u]. \tag{11.3}$$

Together with the correct initial trace this implies (11.2).

Expanded proof (editorial distribution calculation).

Integrate separately on $x < s(t)$ and $x > s(t)$ on a compact time strip. The singular part of u_t is $-s'[u]\delta_{x=s(t)}$; that of $F(u)_x$ is $[F]\delta_{x=s(t)}$. All regular terms vanish by the classical equations. Arbitrary vector tests on the curve force every component of $[F]-s'[u]$ to vanish. Integrating from $t = \delta$ and sending $\delta \downarrow 0$ supplies the initial term if the trace converges in L^1_{loc}. Values assigned on the curve itself do not change any integral. □

11.1.2 Physical models and characteristic speeds

For the elastic p-system use strain a and velocity b:

$$a_t - b_x = 0, \qquad b_t - p(a)_x = 0, \qquad A = DF = \begin{pmatrix} 0 & -1 \\ -p'(a) & 0 \end{pmatrix}.$$

This follows from $a = y_x$, $b = y_t$ for $y_{tt} - (p(y_x))_x = 0$. If $p' > 0$, the speeds are $\lambda_1 = -c$, $\lambda_2 = c$, $c = \sqrt{p'}$, with right vectors $(1, c)$ and $(1, -c)$. For shallow water with gravity scaled to one,

$$h_t + q_x = 0, \qquad q_t + (q^2/h + h^2/2)_x = 0, \qquad h > 0. \tag{11.4}$$

The speeds are $v \mp \sqrt{h}$, $v = q/h$. The strict gap $2\sqrt{h}$ closes at vacuum. For Euler, the conserved variables are $(\rho, \rho v, \rho E)$, $E = e + v^2/2$, and the flux is $(\rho v, \rho v^2 + p, (\rho E + p)v)$. On $\rho > 0$, smooth solutions obey

$$\rho_t + v\rho_x + \rho v_x = 0, \quad v_t + vv_x + \rho^{-1}p_x = 0, \quad e_t + ve_x + (p/\rho)v_x = 0.$$

Subtracting the kinetic-energy balance from total energy gives the last equation. Their matrix is $vI + C$, where

$$C = \begin{pmatrix} 0 & \rho & 0 \\ p_\rho/\rho & 0 & p_e/\rho \\ 0 & p/\rho & 0 \end{pmatrix}, \qquad c^2 = p_\rho + pp_e/\rho^2.$$

Its characteristic polynomial is $\det(\mu I - C) = \mu(\mu^2 - c^2)$. Thus $c^2 > 0$ gives speeds $v-c, v, v+c$. Evans's assumptions $p, p_\rho, p_e > 0$ are sufficient; division by ρ requires positivity.

同一个激波速度

若某分量 $[u^i] = 0$，则须有 $[F^i] = 0$；不能在该分量中作除法。所有非零分量给出的 $[F^i]/[u^i]$ 必须相同，故系统中并非任意两状态都能接激波。

本章的应力符号

这里 $p(a)$ 随应变增加，故 $p' > 0$。气体的 Lagrange 比容形式常采用另一种符号约定，不能把那里的压强导数符号直接搬来。

11.1.3 Strict hyperbolicity and smooth eigenframes

Definition 11.3 (Source §11.1.2)
The system $u_t + B(u)u_x = 0$ is strictly hyperbolic if $B(z)$ has real distinct eigenvalues $\lambda_1 < \cdots < \lambda_m$. Write $Br_k = \lambda_k r_k$, $l_k B = \lambda_k l_k$ and choose $l_j \cdot r_k = \delta_{jk}$. Right eigenvectors are unit when discussing arc length; the dual left eigenvectors need not also be unit.

For $j \neq k$, $(\lambda_j - \lambda_k) l_j \cdot r_k = 0$. The right eigenvectors form a basis, so $l_k \cdot r_k \neq 0$ before normalization. A smooth inviscid traveling wave $V(x - \sigma t)$ satisfies $(B(V) - \sigma I)V' = 0$. Where $V' \neq 0$, its speed must be an eigenvalue along its path. A viscous profile instead uses the scaled variable $(x - \sigma t)/\varepsilon$.

Theorem 11.4 (Coordinate invariance; source §11.1.2, Theorem 1)
Let H be a C^2 diffeomorphism of state domains. For smooth u, $\tilde{u} = H(u)$ satisfies a system with matrix

$$\widetilde{B}(H(z)) = DH(z)B(z)DH(z)^{-1}.$$

Strict hyperbolicity and genuine nonlinearity are preserved. Conservation form need not be preserved.

Expanded proof (chain rule and similarity).

Multiply the PDE by $DH(u)$ and use $\tilde{u}_x = DH(u)u_x$. Similarity preserves eigenvalues. Right vectors become DHr_k; left vectors become $l_k DH^{-1}$. Moreover $D_{\tilde{z}}\widetilde{\lambda}_k \cdot (DHr_k) = D_z \lambda_k \cdot r_k$. Rescaling a vector by a nonzero factor preserves whether this expression is zero. This proof uses a classical chain rule. Through a jump, the corresponding product of a discontinuous coefficient and a measure is not justified. □

Theorem 11.5 (Eigenframe dependence; source §11.1.2, Theorem 2)
A smooth strictly hyperbolic matrix field has smooth ordered eigenvalues and local smooth eigenframes. On state domain $\mathbb{R}^m$, unit eigenvectors can be chosen globally and smoothly. On a general open state domain the eigenframe assertion made here is local.

Expanded proof (including the global orientation step).

At a fixed state take $Br_0 = \lambda_0 r_0$, $|r_0| = 1$. Apply the implicit function theorem to $(B(z)r - \lambda r, (|r|^2 - 1)/2)$. The kernel of its derivative in (r, λ) satisfies $(B - \lambda_0 I)h - \mu r_0 = 0$, $r_0 \cdot h = 0$. Multiplication by l_0 gives $\mu = 0$; simplicity gives $h = cr_0$, then $h = 0$. The square derivative is invertible. Ordered eigenvalues agree on overlaps, so they are global.

局部与全局特征向量

隐函数定理先给局部选择。一般有孔状态域的特征直线丛可能无法连续定向；本章波曲线和 Riemann 拼接只需要一个小邻域。

For global eigenvectors on $\mathbb{R}^m$, form the smooth projection

$$P_k(z) = \prod_{j \neq k} \frac{B(z) - \lambda_j(z) I}{\lambda_k(z) - \lambda_j(z)}.$$

Put $P(s) = P_k(sz)$ and solve $R_s = [P_s, P]R$, $R(0) = r_k(0)$, $0 \le s \le 1$. Differentiating $P^2 = P$ gives $PP_sP = 0$ and $[[P_s, P], P] = P_s$. If $U_s = [P_s, P]U$, $U(0) = I$, then $P(s)U(s) = U(s)P(0)$: subtract the two sides, differentiate, and use this commutator identity and uniqueness for a homogeneous linear ODE. Hence R remains in the range of P, and never vanishes since U is invertible. Smooth dependence on z and normalization at $s = 1$ give a global unit vector. Apply the same construction to the transpose and rescale the left vectors to be dual. This radial construction uses the contractibility of $\mathbb{R}^m$. □

11.2 Riemann's Problem

The initial data are u_l for $x < 0$ and u_r for $x > 0$. All constructions below take place in a sufficiently small strictly hyperbolic neighborhood of a fixed state. Distinguish the state-space curves R_k, S_k from their space-time realizations: the former parameterize possible states, whereas the latter have slopes x/t.

11.2.1 Simple waves and genuine nonlinearity

Seek $u = V(w(x,t))$ with $V' = r_k(V)$. Substitution gives

$$u_t + DF(u)u_x = r_k(V)\{w_t + \lambda_k(V(w))w_x\}.$$

Thus a smooth scalar solution of $w_t + f_k(w)_x = 0$, where $f_k'(s) = \lambda_k(V(s))$, produces a smooth system solution. The integral curve V through z is the k-rarefaction curve $R_k(z)$.

Definition 11.6 (Source §11.2.1)
The k-field is genuinely nonlinear if $\gamma_k = D\lambda_k \cdot r_k \neq 0$ throughout the neighborhood; it is linearly degenerate if $\gamma_k \equiv 0$ there.

For a genuinely nonlinear field orient the unit vector so that $\gamma_k > 0$. Then $f_k'' = \gamma_k(V) > 0$ locally. Reversing the orientation changes the parameter but not the geometric wave.

Theorem 11.7 (Centered rarefaction; source §11.2.2, Theorem 1)
Suppose the k-field is genuinely nonlinear and $u_r = V(s_r)$, $u_l = V(s_l)$ with $s_l < s_r$. On this short curve let $L(s) = \lambda_k(V(s))$. Then the

标量化的适用范围

简单波代入使用链式法则。连续稀疏波可逐片验证再拼接；若标量 w 跳跃，$V(w)$ 未必满足系统 Rankine–Hugoniot 条件，不能直接搬用标量激波。

continuous integral solution is

$$u(x,t) = \begin{cases} u_l, & x/t \le L(s_l), \\ V(L^{-1}(x/t)), & L(s_l) < x/t < L(s_r), \\ u_r, & x/t \ge L(s_r). \end{cases} \tag{11.5}$$

Expanded proof (PDE, interfaces, and the initial trace).

Since $L' = \gamma_k > 0$, its inverse exists and is smooth. In the fan, $U(\xi) = V(L^{-1}(\xi))$ satisfies $(DF(U) - \xi I)U' = 0$, so $u_t + DF(u)u_x = 0$ for $t > 0$. On a strip $t \ge \delta > 0$ the profile is piecewise C^1 and continuous at both fan edges; therefore there is no interface measure in its first derivatives. The fan and all differing initial states occupy an interval of length at most Ct near the origin. Boundedness gives $\|u(\cdot,t) - g\|_{L^1(K)} \le C_K t$ for each compact K. Integration by parts above δ followed by $\delta \downarrow 0$ proves the initial weak identity. □

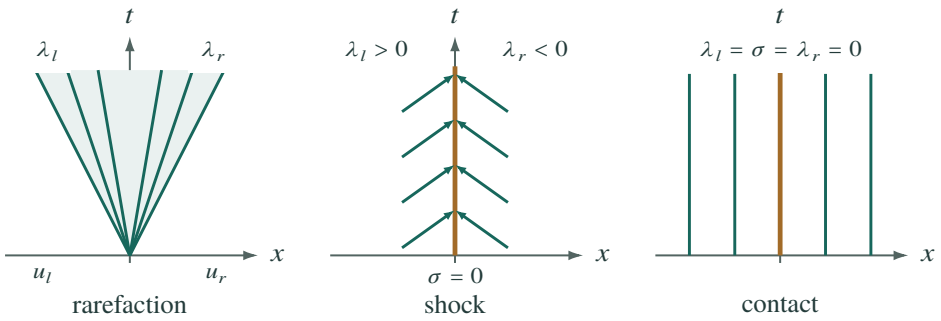


Diagram T11.1. Wave patterns in physical space The left panel is a centered rarefaction: characteristic speeds increase from λ_l to λ_r. The middle panel shows a compressive shock with $\lambda_r < \sigma < \lambda_l$; forward characteristics terminate on it. The right panel is a contact, with equal characteristic and interface speed. Only slopes and incidence are represented; these are schematic panels for a single characteristic family.

11.2.2 The Hugoniot locus and its second-order contact

The shock set at z consists of states v for which $F(v) - F(z) = \sigma(v - z)$ for some common speed σ.

Theorem 11.8 (Local shock curves; source §11.2.3, Theorem 2)
For $F \in C^3$ strictly hyperbolic near z, the shock set is locally the union of m curves $S_k(z)$. In a signed unit-speed parameter a, choose $S_k(0;z) = z$, $\partial_a S_k(0;z) = r_k(z)$. Writing $v = S_k(a;z)$, the shock

speed extends smoothly along this branch to $\sigma_k(0;z) = \lambda_k(z)$, and

$$\partial_a \sigma_k(0;z) = \tfrac{1}{2}\gamma_k(z), \qquad \sigma_k(a;z) = \frac{\lambda_k(v) + \lambda_k(z)}{2} + O(a^2). \tag{11.6}$$

The shock and rarefaction curves have equal second derivatives at zero when both use oriented arc length.

Expanded proof (construction and all Taylor coefficients).

Define $\overline{A}(v,z) = \int_0^1 DF(z+\theta(v-z))\,d\theta$. The jump equation becomes $(\overline{A}(v,z) - \sigma I)(v-z) = 0$. Near $v = z$ this averaged matrix is strictly hyperbolic; its real simple eigenvalues and dual eigenvectors are smooth. Reality follows by continuation of the real simple roots of the characteristic polynomial, and the spectral gaps persist after shrinking the ball. Let these left vectors be $\bar{l}_j(v,z)$. The $m-1$ equations

$$\bar{l}_j(v,z)\cdot(v-z) = 0, \qquad j \neq k,$$

have derivative at $v = z$ with independent rows $l_j(z)$. The implicit function theorem gives a curve tangent to $r_k(z)$. Every nearby nontrivial jump vector is an eigenvector of the averaged matrix, hence belongs to exactly one of these branches. The speed on the branch is its kth eigenvalue; at the common base point there are m distinct branch limits, not a single joint speed function.

Set $r = r_k(z)$, $A = DF(z)$, $\lambda = \lambda_k(z)$ and $H = D^2F(z)[r,r]$. Twice differentiating $F(S(a)) - F(z) = \sigma(a)(S(a)-z)$ at zero gives

$$(A - \lambda I)S''(0) = 2\sigma'(0)r - H.$$

Differentiate $DF\,r_k = \lambda_k r_k$ in direction r:

$$(A - \lambda I)Dr_k(z)r = \gamma_k(z)r - H.$$

Left multiplication by l_k yields $2\sigma'(0) = \gamma_k$. Subtracting the equations shows that $S''(0) - Dr_k r$ is parallel to r. Both terms are orthogonal to r, because both curves have unit speed; hence they are equal. Taylor's formula now gives $\lambda_k(S(a)) = \lambda + \gamma_k a + O(a^2)$ and the stated speed estimate. Local equivalence $|S(a)-z| \asymp |a|$ also permits the remainder to be written $O(|v-z|^2)$. □

Theorem 11.9 (Contacts; source §11.2.3, Theorem 3)
If the k-field is linearly degenerate on a neighborhood, its local shock and rarefaction curves coincide and their speed is $\lambda_k(z) = \lambda_k(v)$. Two such states are joined by a contact at that speed.

激波速度不是全域函数

每一条 S_k 上的速度可光滑延拓到基点，但不同分支极限是不同的 $\lambda_k(z)$。不能在所有分支的交点假定一个共同的光滑 $\sigma(v,z)$。

Expanded proof (inclusion and local equality).

Along $V' = r_k(V)$, $\frac{d}{ds}\lambda_k(V) = 0$. Integration gives $F(V(s)) - F(z) = \int_0^s \lambda_k(z)V'(a)\,da = \lambda_k(z)(V(s) - z)$. Thus the rarefaction curve lies in the shock set. Its tangent places it on branch k; the implicit-function construction gives the unique local curve on that branch. The two germs coincide, and (11.3) gives the contact solution. This is a local equality, not a statement about arbitrary distant branches of the shock set. □

11.2.3 Admissible shocks and the local solver

For a genuinely nonlinear field with $\gamma_k > 0$, (11.6) gives for small $a < 0$

$$\lambda_k(S_k(a;z)) < \sigma_k(a;z) < \lambda_k(z).$$

These are the compressive Lax shocks. For $a > 0$ the inequalities reverse and give expansion shocks, which are rejected. Spectral separation also gives $\lambda_{k-1}(z) < \sigma_k < \lambda_{k+1}(S_k(a;z))$ when the neighboring indices exist. The strict inequalities follow because the leading gaps have size $|\gamma_k a|/2$, whereas the error has size $O(a^2)$.

Define $T_k(a;z)$ as the rarefaction curve for $a \geq 0$ and the shock curve for $a \leq 0$, each with compatible oriented parameters. For a linearly degenerate field use its contact curve for both signs. The common value and tangent, as well as smooth base-state dependence, make T_k jointly C^1 near $(0,z)$; the second-order contact proved above gives additional matching in the wave parameter.

Theorem 11.10 (Local Riemann solver; source §11.2.4, Theorem 4)
If each field is genuinely nonlinear or linearly degenerate on a neighborhood of u_l, every sufficiently close u_r can be joined to u_l by at most m elementary waves, ordered by family. The resulting integral solution is self-similar. Its small wave-strength vector is unique within this construction.

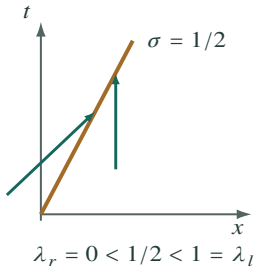
Expanded proof (state matching, speed ordering, weak gluing).

Let $z_0 = u_l$ and $z_k = T_k(a_k; z_{k-1})$. The endpoint map $\mathscr{E}(a) = z_m$ is C^1, $\mathscr{E}(0) = u_l$, and

$$D\mathscr{E}(0) = (r_1(u_l), \dots, r_m(u_l)).$$

This is invertible. The inverse function theorem gives a unique small a with $\mathscr{E}(a) = u_r$. Choose disjoint intervals around the ordered numbers $\lambda_k(u_l)$. For states and strengths sufficiently small, every fan edge, shock speed, and contact speed in family k remains in its own interval. Consequently the elementary waves do not overlap in $\xi = x/t$, and constant

图 T11.M：平均速度与压缩



本图取 Burgers 通量 $f(u) = u^2/2$、$u_l = 1, u_r = 0$。激波速度恰为 $1/2$，两侧速度为 1 和 0；一般系统的平均公式只在小振幅下成立到二阶余项。

intermediate states fill the gaps. Integrate on each smooth sector: continuous fan boundaries have no jump measure, and each shock or contact satisfies (11.3). All interior interface terms cancel. The difference from the initial step is bounded and confined to $|x| \le Ct$, giving the initial trace as in the rarefaction proof. The inverse theorem proves uniqueness of the small parameters; it does not prove uniqueness among all bounded weak or entropy solutions of the system. □

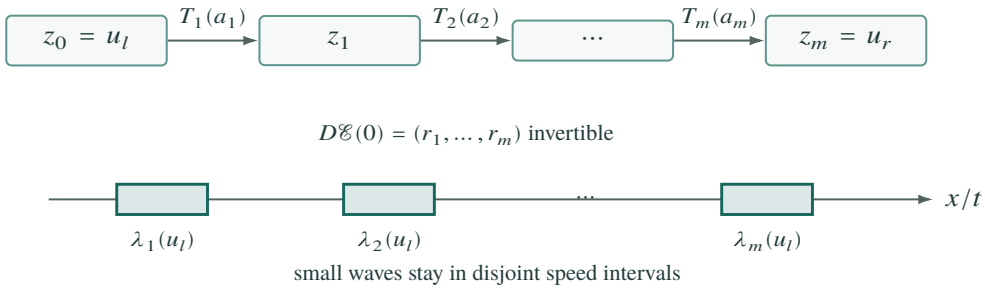


Diagram T11.2. Matching states before placing waves The endpoint map composes the family curves in state space. Its derivative at zero consists of the independent eigenvectors. Disjoint speed intervals then place those waves in increasing order in x/t. The unique small strength vector is a local finite-dimensional conclusion; no general system uniqueness theorem is inferred.

波曲线的方向

本稿始终写右状态 $z_k \in T_k(z_{k-1})$。原书 §11.2.4 的式 (43) 与随后 (45) 的左右方向写法不一致；这里以构造式 (45) 为准，明确每次从左状态走向右状态。

11.3 Systems of Two Conservation Laws

11.3.1 Local Riemann coordinates

In this section $m = 2$. Evans calls w_i an ith Riemann invariant when $Dw_i \cdot r_i = 0$: it is constant on the state-space curve R_i. A useful coordinate must also have $Dw_i \neq 0$.

Lemma 11.11 (Existence of local invariant coordinates; editorial detail)
Near a strictly hyperbolic state there are smooth w_1, w_2 satisfying $Dw_i \cdot r_i = 0$ and $\det D(w_1, w_2) \neq 0$.

Expanded proof (flow-box construction).

Choose a short curve $\Gamma(a)$ transverse to r_i at the base point. Let $X(s,a)$ solve $X_s = r_i(X)$, $X(0,a) = \Gamma(a)$. At the base point the two columns r_i, Γ' are independent, so the inverse function theorem makes (s,a) local coordinates. Define $w_i(X(s,a)) = a$. Then $Dw_i r_i = 0$ and $Dw_i \neq 0$. In two dimensions the annihilator of r_i is spanned by $l_j, j \neq i$. Thus Dw_1 and Dw_2 are nonzero multiples of independent left eigenvectors. After intersecting their neighborhoods, the two functions together are coordinates. □

Theorem 11.12 (Diagonal form; source §11.3.1, Theorem 1)
For a smooth solution, set $v_i = w_i(u)$ and express $u = z(v_1, v_2)$. Then

$$(v_1)_t + \lambda_2(v)(v_1)_x = 0, \qquad (v_2)_t + \lambda_1(v)(v_2)_x = 0. \tag{11.7}$$

Here $\lambda_i(v)$ means $\lambda_i(z(v))$.

Expanded proof (why the indices cross).

Because Dw_i is parallel to l_j, $j \neq i$, $Dw_i(-DF + \lambda_j I)u_x = 0$. The chain rule gives the displayed transport equation. Along a curve $X' = \lambda_j(u(X,t))$ its v_i value is constant. Also z_{v_j} is a nonzero multiple of r_i for $i \neq j$: it lies in the kernel of Dw_i and cannot be zero since Dz is invertible. Therefore genuine nonlinearity of family i is equivalent to $\partial_{v_j}\lambda_i \neq 0$. The transport speeds still depend on both invariants, so the diagonal equations are coupled, not independent scalar equations. □

For barotropic gas in conserved variables (ρ, q),

$$F = (q, q^2/\rho + p(\rho)), \qquad \rho > 0, \quad p' > 0.$$

In primitive variables (ρ, v), $v = q/\rho$, the matrix is $\left(\begin{smallmatrix} v & \rho \\ p'/\rho & v \end{smallmatrix}\right)$, with speeds $v \mp c$, $c = \sqrt{p'}$. Put

$$I(\rho) = \int_{\rho_*}^{\rho} \frac{c(s)}{s}\, ds, \qquad w_1 = I(\rho) + v, \quad w_2 = I(\rho) - v. \tag{11.8}$$

The Jacobian determinant in (ρ, v) is $-2c/\rho \neq 0$. Direct substitution of $\rho_t + v\rho_x + \rho v_x = 0$ and $v_t + vv_x + (c^2/\rho)\rho_x = 0$ proves $D_+ w_1 = 0$, $D_- w_2 = 0$, where $D_\pm = \partial_t + (v \pm c)\partial_x$. These equations use only the barotropic mass–momentum model. It is not necessary to impose the full Euler energy equation with constant internal energy.

11.3.2 Compression and a weighted Riccati equation

We give a version of source §11.3.2, Theorem 2 with its uniformity assumptions explicit. Suppose the characteristic variables exist on an open set containing the closed rectangle $K = [A_1, B_1] \times [A_2, B_2]$. Assume the coefficients are C^2 there,

$$\lambda_2 - \lambda_1 \geq d > 0, \qquad \partial_{v_1}\lambda_2 \geq c_0 > 0, \qquad \partial_{v_2}\lambda_1 \geq c_0 > 0 \quad \text{on } K. \tag{11.9}$$

The signs can be obtained from genuine nonlinearity by suitable orientations of the two invariant coordinates on a connected chart.

两种“不变”曲线

按 Evans 的编号，w_1 在状态空间沿 R_1 不变，但 $w_1(u(x,t))$ 沿时空的 λ_2 特征线不变。这两个陈述分别使用右特征向量与另一族左特征向量，并不矛盾。

远离真空的条件

当 $\rho \downarrow 0$，c/ρ 可能发散，特征间隙也可能消失。局部坐标可逆不保证整个解的值域始终留在同一张坐标图中；下面的爆破定理显式规定紧矩形值域。

Theorem 11.13 (Finite-time loss of smoothness)
Suppose the smooth initial invariants range in K, and a classical solution remains in this chart with range in K while it exists. If either initial derivative $(v_1)_x$ or $(v_2)_x$ is negative somewhere, it cannot remain a classical solution for all positive time. More precisely a weighted derivative diverges to $-\infty$ in a finite bounded time unless the assumed smooth regime ends sooner.

爆破的是梯度

输运使 v_i 保持初值范围，而 $e^H(v_1)_x$ 满足负二次 ODE。证明使用的是系数的一致正下界；只有逐点正性而没有紧值域控制，不能据此断言积分必在有限时刻达到临界值。

Expanded proof (removing the mixed derivative term).

Set $a = (v_1)_x$, $b = (v_2)_x$ and $D_2 = \partial_t + \lambda_2\partial_x$. Differentiating the first equation in (11.7) gives

$$D_2 a = -(\partial_1\lambda_2)a^2 - (\partial_2\lambda_2)ab, \qquad D_2 v_2 = (\lambda_2 - \lambda_1)b.$$

Choose H with

$$H(v_1, v_2) = \int_{A_2}^{v_2} \frac{\partial_2\lambda_2(v_1, s)}{\lambda_2(v_1, s) - \lambda_1(v_1, s)}\, ds.$$

Along a λ_2 characteristic, v_1 is constant, so $D_2 H = (\partial_2\lambda_2)b$. For $y = e^H a$ the mixed term cancels:

$$\frac{dy}{dt} = -k(t)y^2, \qquad k = e^{-H}\partial_1\lambda_2 \geq \kappa > 0. \tag{11.10}$$

The lower bound follows from compactness of K and (11.9), not merely pointwise positivity. If $y(0) < 0$, uniqueness preserves its sign before blow-up, and

$$\frac{1}{y(t)} = \frac{1}{y(0)} + \int_0^t k(s)\, ds.$$

The denominator reaches zero no later than $T_* = -1/(\kappa y(0))$. Thus a globally finite derivative is impossible. For b, use $D_1 = \partial_t + \lambda_1\partial_x$ and a weight J satisfying $\partial_1 J = \partial_1\lambda_1/(\lambda_1 - \lambda_2)$; then $D_1(e^J b) = -e^{-J}\partial_2\lambda_1(e^J b)^2$.

On a smooth solution confined to the chart, the characteristic speeds are bounded and locally Lipschitz in space, so the characteristics exist on every finite smooth time strip. Each invariant is transported from its initial range; this explains invariance of the rectangle provided the chart covers that entire rectangle. It does not supply a missing global inverse chart. Smooth compactly supported perturbations of a constant state have bounded invariant ranges. Any nonconstant one of these invariants, with equal limits at $\pm\infty$, has a negative derivative somewhere, so the criterion applies under the stated chart and uniformity assumptions. □

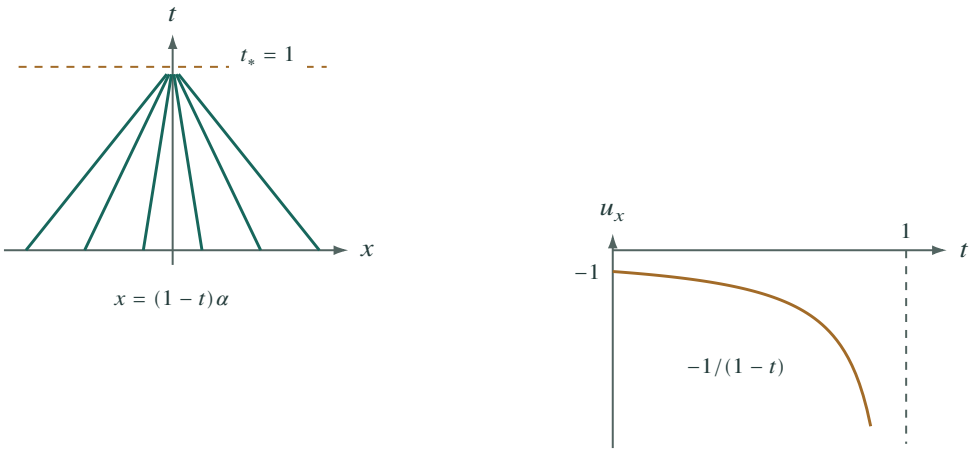


Diagram T11.3. Compression reaches a vertical slope An exact simple-wave example is the scalar Burgers characteristic map $x = \alpha + t\,g(\alpha)$ for $g(\alpha) = -\alpha$ on a local interval. Its Jacobian is $1 - t$, and $u_x = -1/(1 - t)$ until $t = 1$. The second panel plots this derivative. For two-field systems, the weight in (11.10) removes the mixed interaction and recovers the same Riccati mechanism; the drawn map is the scalar model.

11.4 Entropy Criteria

11.4.1 Viscous profiles and the limits of dimension counting

For identity viscosity, $u_t^\varepsilon + F(u^\varepsilon)_x = \varepsilon u_{xx}^\varepsilon$. The ansatz $u^\varepsilon = V((x-\sigma t)/\varepsilon)$ gives

$$\begin{aligned} V' &= F(V) - F(u_l) - \sigma(V - u_l) = G(V;\sigma), \\ V(-\infty) &= u_l, \qquad V(+\infty) = u_r. \end{aligned} \tag{11.11}$$

Here integration used $V'(-\infty) = 0$; the right limit then forces Rankine–Hugoniot. For an autonomous smooth ODE, convergence of V to an equilibrium also implies $V' \to 0$.

We use the following *external finite-dimensional theorem*, not proved here: a sufficiently smooth autonomous vector field with a spectral splitting into center, stable, and unstable subspaces has a local center manifold of any fixed finite differentiability allowed by the field. It is tangent to the center space and contains every complete orbit staying in a sufficiently small neighborhood, in particular all nearby equilibria. Parameters can be included as variables with zero derivative. The conclusion is local; it does not assert that a distant excursion lies on the manifold.

稳定维数不等于轨道存在

线性化特征值是 $\lambda_j(u_l) - \sigma$。某个方向不稳定只说明允许离开平衡点，不能单凭维数判断一条轨道必到达指定的右状态；一般系统须另用中心流形等工具。

Theorem 11.14 (Small local viscous profiles)
This is a local version of source §11.4.1, Theorem 1. Assume F is smooth and the k-field is genuinely nonlinear near u_l. For a sufficiently small nonzero k-shock state u_r, a profile whose entire image

> *stays near u_l exists if and only if it is on the compressive branch $S_k^-(u_l)$. The proof below uses the stated external center-manifold theorem.*

Expanded proof (center-manifold reduction).

Fix u_l, orient $r = r_k(u_l)$ so $\gamma = D\lambda_k(u_l)r > 0$, and write $\beta = \sigma - \lambda_k(u_l)$. Adjoin $\beta' = 0$ to (11.11). At $(V, \beta) = (u_l, 0)$ the center space has dimension two, spanned by $(r, 0)$ and $(0, 1)$; all other eigenvalues are separated from zero by strict hyperbolicity. A local center manifold can be parameterized by $a = l_k(u_l)(V - u_l)$ and β:

$$V = u_l + ar + W(a, \beta), \quad l_k W = 0, \quad W(0, \beta) = 0, \quad W = O(a^2 + |a\beta|).$$

The equality $W(0, \beta) = 0$ follows because $V = u_l$ is an equilibrium for every small β. Project the vector field by l_k. Taylor expansion, $l_k(DF - \lambda_k I) = 0$, and $l_k r = 1$ give

$$a' = g(a, \beta) = -\beta a + \tfrac{1}{2}\gamma a^2 + O(|a|(|a| + |\beta|)^2).$$

The coefficient identity $l_k D^2 F[r, r] = \gamma$ follows by differentiating the eigenvalue equation. The nearby right equilibrium lies on the manifold. Write its coordinate $a_r \neq 0$; by the shock speed expansion, $\beta = \gamma a_r/2 + O(a_r^2)$. For this fixed β, both 0 and a_r are roots of g. Divide smoothly first by a and then by $a - a_r$, using the integral form of the mean value theorem. Uniformly for a between the roots,

$$g(a, \beta) = a(a - a_r)h(a, \beta), \qquad h = \gamma/2 + O(|a_r|) > 0.$$

If $a_r < 0$, the vector field is negative between a_r and 0. The solution through any interior point decreases, stays between the equilibria, and converges to 0 as $s \to -\infty$ and to a_r as $s \to +\infty$. The simple roots make the times to both endpoints infinite (the separated integral has logarithmic divergences). Lifting gives the desired profile, with its full image small. If $a_r > 0$, the field is negative on $(0, a_r)$ and cannot connect 0 to a_r forward in s. Every complete sufficiently small profile lies on the center manifold, so this also proves the local necessity. The equivalence $a_r < 0 \iff u_r \in S_k^-(u_l)$ follows from (11.6). No assertion about profiles making large excursions is obtained from this local argument. □

For a non-genuinely nonlinear branch, the Liu criterion compares shock speeds along the full intervening Hugoniot arc:

$$\sigma(z, u_l) \geq \sigma(u_r, u_l) \quad \text{for intermediate states } z \in S_k(u_l).$$

The source writes a strict version for nontrivial intermediate states. The local general existence theory requires additional hypotheses and is not

proved by the preceding genuinely nonlinear reduction. In the scalar profile calculation below, strictness has a precise meaning: it excludes an extra equilibrium that would block passage.

11.4.2 A complete profile test for the viscous elastic system

Use viscosity only in the velocity equation: $a_t - b_x = 0$, $b_t - p(a)_x = \varepsilon b_{xx}$. For end strains $a_l < a_r$ choose

$$\sigma = \sqrt{\frac{p(a_r) - p(a_l)}{a_r - a_l}} > 0, \qquad b_r = b_l - \sigma(a_r - a_l), \qquad p' > 0.$$

Theorem 11.15 (Source §11.4.1, elastic example)
A traveling profile joining these states exists if and only if

$$p(a) - p(a_l) > \sigma^2(a - a_l), \qquad a_l < a < a_r. \tag{11.12}$$

It is unique up to translation among such monotone profiles.

Expanded proof (separation and infinite endpoint time).

The first integrated equation gives $B = b_l - \sigma(A - a_l)$. The second yields

$$A' = g(A) = \frac{p(A) - p(a_l) - \sigma^2(A - a_l)}{\sigma}.$$

Thus $g(a_l) = g(a_r) = 0$. If $g > 0$ in the open interval, the inverse of $s - s_0 = \int_{A_0}^{A} d\zeta / g(\zeta)$ is a strictly increasing profile. Since g is C^1 and vanishes at each endpoint, $|g(a)| \le C|a - a_l|$ near a_l and similarly near a_r; the endpoint integrals diverge, even if the roots are not simple. The inverse is defined for all real s with the required limits. Conversely any connecting scalar orbit must pass every intermediate value. Uniqueness for a scalar Lipschitz ODE prevents passage through an interior zero and prevents a change of direction. Hence $g > 0$ throughout the interval. Separation also proves translation uniqueness. Condition (11.12) is precisely strict decrease of the secant speed from a_l to an intermediate point relative to the final positive speed. This proof concerns the stated partial viscosity, not an unmentioned change of viscosity matrix. □

11.4.3 Entropy pairs, symmetrizers, and viscous limits

Definition 11.16 (Source §11.4.2)
An entropy pair (η, Q) consists of C^2 functions with η convex and $DQ = D\eta\, DF$. An integral solution is an entropy solution if for every

such pair and $0 \le \varphi \in C_c^\infty(\mathbb{R} \times (0,\infty))$,

$$\iint [\eta(u)\varphi_t + Q(u)\varphi_x]\,dx\,dt \ge 0. \tag{11.13}$$

Thus the distribution $\eta(u)_t + Q(u)_x$ is nonpositive.

For a smooth solution the entropy balance is equality. At a shock it reduces to $[Q] - \sigma[\eta] \le 0$ by the same interface calculation as before. The sign refers to convex mathematical entropy, often the negative of physical entropy.

Lemma 11.17 (Entropy compatibility; editorial supplement)
Locally, $D\eta\,DF$ is the gradient of a flux if and only if $D^2\eta\,DF$ is symmetric. If $D^2\eta$ is positive definite, DF is similar to a symmetric matrix and has real eigenvalues. This does not force distinctness.

Expanded proof (integrability and symmetrization).

For $b_j = \sum_a \eta_a F_j^a$, differentiate with respect to z_i. In $\partial_i b_j - \partial_j b_i$ the terms involving F_{ij}^a cancel. The remaining expression is the skew part of $D^2\eta DF$. A closed C^1 one-form has a potential on a small ball: integrate along radial segments and differentiate, using equality of the cross partials. This proves local sufficiency; global sufficiency requires, for example, a simply connected domain. If $S = D^2\eta > 0$, then $S^{1/2}DFS^{-1/2}$ is symmetric because $SDF = (DF)^T S$. It is similar to DF and hence has the same real spectrum. □

For the elastic system, a strictly convex pair is $\eta(a,b) = b^2/2 + P(a)$, $P' = p$, and $Q = -bp(a)$. Indeed $D^2\eta = \operatorname{diag}(p', 1) > 0$.

Theorem 11.18 (Viscous limit; source §11.4.2, Theorem 2)
Suppose smooth identity-viscosity solutions with initial data g are uniformly bounded with range in a fixed compact state set and converge a.e. to u on $\mathbb{R} \times (0,\infty)$. Then u is an integral solution and satisfies (11.13) for all smooth convex entropy pairs. This is conditional on the convergence; it is not a compactness or existence theorem for general systems.

Expanded proof (moving derivatives onto the compact test).

The chain rule yields

$$\eta(u^\varepsilon)_t + Q(u^\varepsilon)_x = \varepsilon\eta(u^\varepsilon)_{xx} - \varepsilon(u_x^\varepsilon)^T D^2\eta(u^\varepsilon)u_x^\varepsilon.$$

The last term is nonpositive. Multiply by $\varphi \ge 0$ and integrate:

$$\iint (\eta(u^\varepsilon)\varphi_t + Q(u^\varepsilon)\varphi_x) \ge -\varepsilon \iint \eta(u^\varepsilon)\varphi_{xx}.$$

系统熵对的相容性

标量情形每个凸 η 都能积分得到 Q；系统须满足超定条件 $D^2\eta\,DF$ 对称。并不是任意凸函数都可充当某个系统的熵，更不能由一个能量不等式直接得到所有解的唯一性。

The error is $O(\varepsilon)$ because the test support and the range are compact. Dominated convergence handles the other terms. For the weak PDE, test the viscous equation directly, including $t = 0$:

$$\iint (u^\varepsilon \cdot \varphi_t + F(u^\varepsilon) \cdot \varphi_x + \varepsilon u^\varepsilon \cdot \varphi_{xx}) + \int g \cdot \varphi(\cdot, 0) = 0.$$

The same argument proves (11.2). No integration over spatial infinity, energy decay assumption, or gradient bound is needed for these compactly supported tests. An arbitrary nonsmooth system entropy is not automatically covered by scalar convolution: that convolution need not preserve the system flux compatibility. □

11.4.4 Scalar entropy and the full contraction argument

Let $m = 1, f \in C^2(\mathbb{R})$, and work in $u \in C([0,\infty); L^1(\mathbb{R})) \cap L^\infty$ with initial trace g. For each $k \in \mathbb{R}$, the Kruzhkov pair is

$$\eta_k(u) = |u - k|, \qquad q_k(u) = \operatorname{sgn}(u - k)(f(u) - f(k)).$$

It follows from the smooth pairs by replacing $|s-k|$ by $\sqrt{(s-k)^2 + \delta^2}$ and taking $Q_\delta(s) = \int_k^s \eta'_\delta(r) f'(r)\, dr$. On a bounded range both components converge uniformly, so entropy inequalities pass to the limit.

Theorem 11.19 (Scalar contraction; source §11.4.3, Theorem 3)
If u, v are scalar entropy solutions in this class with data g, h, then for all $0 \le s \le t$,

$$\|u(\cdot, t) - v(\cdot, t)\|_1 \le \|u(\cdot, s) - v(\cdot, s)\|_1. \tag{11.14}$$

In particular, the solution with a prescribed initial trace is unique. No convexity of f is required.

Expanded proof (doubling, diagonal limit, and endpoint times).

For fixed (y, s) use $k = v(y, s)$ in the entropy inequality for u, and for fixed (x, t) use $k = u(x, t)$ in that for v. The pair $q(a, b) = \operatorname{sgn}(a - b)(f(a) - f(b))$ is symmetric in a, b. Add and integrate the inequalities to get, for nonnegative smooth compactly supported $\zeta(x, t, y, s)$ away from $t = 0, s = 0$,

$$\iiiint \{|u - v|(\partial_t + \partial_s)\zeta + q(u, v)(\partial_x + \partial_y)\zeta\} \ge 0.$$

Fubini is legitimate because all variables are bounded on the compact support; the inequalities hold for every constant k and smooth test. Take

$$\zeta = \rho_\delta(x - y)\rho_\delta(t - s)\phi\big((x + y)/2, (t + s)/2\big), \qquad \phi \ge 0.$$

For small δ the support remains away from both initial times. The difference-kernel derivatives cancel under the summed derivatives; the derivatives of ϕ have coefficient one. Use mean and difference variables; their constant positive Jacobian can be divided out. Translations are continuous in L^1 on compact space-time sets. The functions $|a-b|$ and $q(a,b)$ are jointly Lipschitz on any bounded square: for fixed b, the piecewise derivative of q is $\pm f'(a)$ and the two pieces meet continuously at $a=b$; then repeat in b. Thus the diagonal limit gives

$$\iint (A\phi_t + B\phi_x) \geq 0, \quad A = |u-v|, \quad B = q(u,v), \quad |B| \leq LA, \tag{11.15}$$

where $L = \max_{|r|\leq M} |f'(r)|$ and M bounds both solutions.

Choose $\phi = \alpha_R(x)\beta(\tau)$, with $0 \leq \alpha_R \leq 1$, $\alpha_R = 1$ on $[-R,R]$, zero outside $[-2R,2R]$, $|\alpha_R'| \leq C/R$, and $\beta \geq 0$ compact in positive time. The flux contribution tends to zero, since its absolute value is at most $(CL/R)\|\beta\|_\infty \int \|u-v\|_1 d\tau$. The time integral is finite by L^1 continuity on compact intervals. Dominated convergence yields $\int \|u-v\|_1 \beta' \geq 0$. Hence the distributional time derivative of this continuous function is nonpositive. Equivalently, approximate the indicator of $[s,t]$ by trapezoidal Lipschitz functions and then mollify; the two end averages tend to the values at s,t by continuity. This proves contraction first for $s>0$, then for $s=0$ by the strong initial traces. Continuity also upgrades an a.e.-time inequality to every time. □

The same Kato inequality gives a local finite-speed estimate:

$$\int_{|x|\leq R} |u(x,t) - v(x,t)|\,dx \leq \int_{|x|\leq R+Lt} |g(x) - h(x)|\,dx.$$

To verify it, use cutoffs of the backward interval $|x| < R + L(t-\tau)$ in (11.15); they satisfy $\phi_\tau + L|\phi_x| \leq 0$ between the two time cutoffs. The bound $|B| \leq LA$ makes the lateral flux nonpositive. Smooth approximation and the strong local traces supply the two end sections. This local version also proves uniqueness for bounded nonintegrable Riemann data with L^1_{loc} time continuity.

初始迹与全时刻唯一性

消黏的分布初值不自动给强 L^1 连续性。本定理把 $C_tL^1_x$ 与强初始迹作为解类条件，再把几乎处处时刻的收缩提升到每个时刻。一般系统没有相同的序结构。

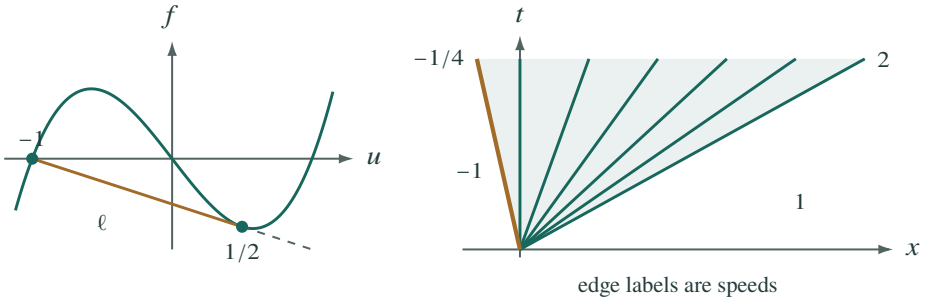


Diagram T11.4. The entropy chord for a nonconvex flux For $f(u) = u^3 - u$, the chord from $u_l = -1$ to $u_* = 1/2$ has slope $-1/4 = f'(1/2)$ and lies below f on that interval. The entropy solution for right state 1 joins this shock to a rarefaction $u = \sqrt{(x/t+1)/3}$ for $-1/4 < x/t < 2$. A single jump from -1 to 1 violates the chord criterion. The exact states and speeds are derived in Problems 14–15.

11.5 Problems

The following are abbreviated descriptions of the 17 source problems. All solutions and additional qualifications are editorial reconstructions.

1. Generalized elastic system

For $F = (-q(b), -p(a))$ with $p', q' > 0$,

$$DF = \begin{pmatrix} 0 & -q'(b) \\ -p'(a) & 0 \end{pmatrix}, \qquad \lambda_{\pm} = \pm\sqrt{p'(a)q'(b)}.$$

They are real and distinct. Corresponding vectors are $(q'(b), -\lambda_{\pm})$; substitution verifies the eigenvalue equation.

2. Shallow water and a change of conserved variables

In (h, q), $DF = \begin{pmatrix} 0 & 1 \\ h - q^2/h^2 & 2q/h \end{pmatrix}$, whose characteristic polynomial is $(\lambda - v)^2 - h$. For smooth $h > 0$, expansion of $q = hv$ and use of $h_t + (hv)_x = 0$ give $v_t + vv_x + h_x = 0$. Hence the alternative smooth system has flux $\widetilde{F}(h, v) = (hv, v^2/2 + h)$ and matrix $\begin{pmatrix} v & h \\ 1 & v \end{pmatrix}$, again with speeds $v \pm \sqrt{h}$. This is equivalence for smooth solutions only. For example, the original jump from $(h, v) = (1, 0)$ to $(2, \sqrt{3}/2)$ has speed $\sigma = \sqrt{3}$ and satisfies both original conservation laws. For the alternative velocity law its flux jump is $11/8$, whereas $\sigma[v] = 3/2$; the jump fails. Nonlinear coordinate changes can therefore alter admissible weak solutions even when both resulting smooth systems happen to have conservation form.

3. Smooth matrices with rotating eigenspaces

For $z \neq 0$ put $\theta = 1/z$ and

$$B(z) = e^{-1/z^2} \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \qquad B(0) = 0.$$

Every derivative is a finite sum of a bounded sine or cosine times $z^{-N} e^{-1/z^2}$, which tends to zero. Induction proves $B \in C^\infty$. The eigenvalues are $\pm e^{-1/z^2}$ for $z \neq 0$, but coincide at zero. The positive eigenline is spanned by $(\cos\theta, \sin\theta)$; the negative line is its perpendicular. Along sequences $\theta = n\pi$ and $\theta = \pi/2 + n\pi$, even the positive line has incompatible limits. Sign choices cannot repair this, so no continuous unit frame extends through zero. This failure concerns loss of strictness.

4. Shallow-water rarefaction curves

In (h, v) use $r_1 = (1, -1/\sqrt{h})$, $r_2 = (1, 1/\sqrt{h})$. Integrating $dv/dh = \mp h^{-1/2}$ gives $2\sqrt{h} + v = C$ on R_1 and $2\sqrt{h} - v = C$ on R_2. The gradient of each displayed invariant annihilates its own vector.

5. Shallow-water shock curves

Write the base state as $(h_0, h_0 v_0)$. Eliminating σ from $\sigma[h] = [hv]$, $\sigma[hv] = [hv^2 + h^2/2]$ gives, for $h \neq h_0$,

$$v = v_0 \pm (h - h_0)\sqrt{\frac{h + h_0}{2hh_0}}, \qquad q = hv_0 \pm (h - h_0)\sqrt{\frac{h(h + h_0)}{2h_0}}. \tag{11.16}$$

One way to see the elimination is to introduce the relative mass flux $j = h(v - \sigma) = h_0(v_0 - \sigma)$; momentum gives $j^2 = (h^2 - h_0^2)/(2(1/h_0 - 1/h)) = hh_0(h + h_0)/2$. The nontrivial branch speeds are $\sigma = v_0 \pm \sqrt{h(h + h_0)/(2h_0)}$. If $h = h_0$, the mass jump forces $q = q_0$, so there is no missing nonzero jump. For $q_0 = 0$, the formula is exactly $q = \pm(h - h_0)h\sqrt{(1/h + 1/h_0)/2}$.

6. Dam-break Riemann data with positive depths

Let $h_l > h_r > 0$ and both initial velocities be zero. The intermediate depth H is the unique root in (h_r, h_l) of

$$2(\sqrt{h_l} - \sqrt{H}) = (H - h_r)\sqrt{\frac{H + h_r}{2Hh_r}} =: V. \tag{11.17}$$

The left member strictly decreases and the right strictly increases: for the latter square it and differentiate $(H - h_r)^2(H + h_r)/(2Hh_r)$, or note both

positive factors $H-h_r$ and $\sqrt{1+h_r/H}$ have combined positive derivative. The endpoint signs are opposite, proving existence and uniqueness. Set $q_i = HV$, $\sigma = HV/(H-h_r)$. With $\xi = x/t$,

$$(h,v) = \begin{cases} (h_l, 0), & \xi \le -\sqrt{h_l}, \\ ((2\sqrt{h_l} - \xi)^2/9,\ 2(\sqrt{h_l} + \xi)/3), & -\sqrt{h_l} < \xi < V - \sqrt{H}, \\ (H, V), & V - \sqrt{H} \le \xi < \sigma, \\ (h_r, 0), & \xi > \sigma. \end{cases}$$

The fan preserves $v + 2\sqrt{h} = 2\sqrt{h_l}$ and has $\xi = v - \sqrt{h}$. Its right edge precedes the shock. Indeed

$$\sigma^2 = \frac{H(H+h_r)}{2h_r} > h_r, \qquad (\sigma - V)^2 = \frac{h_r(H+h_r)}{2H} < H.$$

Thus $\sqrt{h_r} < \sigma < V + \sqrt{H}$, the Lax inequalities for a 2-shock; also $V - \sqrt{H} < \sigma$. The fan is continuous, the shock satisfies both vector jumps, and the $O(t)$ disturbed interval proves the initial trace. For all smooth entropy pairs the shock inequality can also be checked without assuming small amplitude. Hold the left state (H, HV) fixed and follow its 2-shock branch toward smaller h. Parameterize by $s = H - h > 0$. Formula (11.16) gives $\sigma(s) = V + \sqrt{h(h+H)/(2H)}$, strictly decreasing in s. For $E(s) = [Q] - \sigma(s)[\eta]$, differentiated Rankine–Hugoniot gives

$$E'(s) = \sigma'(s)\{D\eta(u_r(s)) \cdot (u_r(s) - u_l) - \eta(u_r(s)) + \eta(u_l)\} \le 0$$

by convexity, and $E(0) = 0$. Thus this is an entropy solution in the system sense, including for large positive depth ratios.

7. Barotropic invariants in the proper variables

For $z = (\rho, q)$, use $r_1 = (1, v-c)$ and $r_2 = (1, v+c)$. The gradients of (11.8) are

$$Dw_1 = ((c-v)/\rho, 1/\rho), \qquad Dw_2 = ((c+v)/\rho, -1/\rho).$$

Their respective products with r_1, r_2 vanish. This verifies the source definition in conserved, rather than merely primitive, variables.

8. Compatibility for the alternative shallow-water system

For $\widetilde{F} = (hv, v^2/2 + h)$ the entropy relation reads $Q_h = v\eta_h + \eta_v$, $Q_v = h\eta_h + v\eta_v$. Equality of their cross derivatives gives $\eta_{vv} = h\eta_{hh}$. This is necessary for a flux; convexity of η is a separate part of being an entropy pair.

水坝问题不必小振幅

局部 Riemann 定理只保证邻近状态。本题通过单调求根、精确跳跃条件及沿激波支的熵耗散，单独处理任意 $h_l > h_r > 0$；不把局部定理扩大到真空或任意大数据。

9. Mechanical entropy for barotropic gas

In conserved variables (ρ, q) put $\eta = q^2/(2\rho) + P(\rho)$ with $P'' = p'/\rho$. A compatible flux is

$$Q = \frac{q^3}{2\rho^2} + qP'(\rho).$$

Direct differentiation gives $DQ = D\eta DF$. If the affine freedom in P is chosen so $\rho P' - P = p$, this is $Q = v(\eta + p)$; without that normalization these expressions differ by a constant multiple of the mass flux q and one must use the compatible expression above. In fact $\rho P' - P - p$ is a constant, so choosing the additive constant in P removes it. The Hessian quadratic form is

$$D^2\eta[(a,b),(a,b)] = \frac{p'(\rho)}{\rho}a^2 + \frac{1}{\rho}(b - va)^2 > 0 \quad ((a,b) \neq 0).$$

Convexity here is in (ρ, q), not asserted in (ρ, v).

10. Gradient flux

For $F = D\phi$ take $\eta = |z|^2/2$ and $Q = z \cdot D\phi - \phi$. The derivative of $z \cdot D\phi$ is $D\phi + z^T D^2\phi$; cancellation leaves $DQ = z^T D^2\phi = D\eta DF$. The Hessian of η is I.

11. Antigradient flux and the generalized elastic energy

For $F = (\phi_2, \phi_1)$ and convex ϕ, $\eta = \phi$, $Q = \phi_1\phi_2$. Componentwise, $D\eta DF = (\phi_1\phi_{12} + \phi_2\phi_{11}, \phi_1\phi_{22} + \phi_2\phi_{12}) = DQ$. For Problem 1 set $P' = p$, $R' = q$, and use

$$\eta(a,b) = P(a) + R(b), \qquad Q(a,b) = -p(a)q(b).$$

Then $D^2\eta = \operatorname{diag}(p', q') > 0$ and the compatibility identity follows directly. This energy is obtained by calculation; the flux itself need not come from a convex antigradient potential with the same sign as in the first part.

12. Several space dimensions

For $F = (F_1, \dots, F_n)$ with $F_j : \Omega \to \mathbb{R}^m$, an entropy pair is a convex η and $Q = (Q_1, \dots, Q_n)$ such that $DQ_j = D\eta DF_j$ for each j. Smooth solutions obey $\eta(u)_t + \operatorname{div} Q(u) = 0$; the entropy inequality for weak solutions is tested against every nonnegative compactly supported scalar space-time function.

13. Nonlinear dielectric Maxwell energy

With $B_t = -\operatorname{curl} E$, $D_t = \operatorname{curl} H$, $E = D_D\eta$, $H = D_B\eta$, the chain rule gives

$$\eta_t = H \cdot (-\operatorname{curl} E) + E \cdot \operatorname{curl} H = -\operatorname{div}(E \times H).$$

The identity $\operatorname{div}(E \times H) = H \cdot \operatorname{curl} E - E \cdot \operatorname{curl} H$ fixes the sign. To verify the algebraic entropy compatibility, the jth fluxes for the state (B, D) have components $F_j^{B_i} = \epsilon_{ijk}E_k$ and $F_j^{D_i} = -\epsilon_{ijk}H_k$. Hence $D\eta\, dF_j = H_i\epsilon_{ijk}dE_k - E_i\epsilon_{ijk}dH_k = d(E \times H)_j$ (summation over i, k). Convexity of the given energy supplies the entropy condition.

14. The entropy chord criterion for a scalar shock

Let $a = u_l$, $b = u_r$, $a \neq b$, and $\sigma = (f(b) - f(a))/(b - a)$. Write $\ell(k) = f(a) + \sigma(k - a)$. For $a < k < b$, the Kruzhkov jump defect is

$$[q_k] - \sigma[\eta_k] = 2(\ell(k) - f(k)).$$

For $b < k < a$, the defect is $2(f(k) - \ell(k))$. Outside the interval it is zero by Rankine–Hugoniot. Thus for an increasing jump entropy requires $f \geq \ell$ on $[a, b]$; for a decreasing jump it requires $f \leq \ell$ on $[b, a]$. This also suffices for every smooth convex entropy: on a bounded interval any such entropy is an affine function plus a positive superposition of $|u - k|$, with weights $\eta''(k)/2$; integrate the corresponding jump inequalities. The affine part contributes zero by the vector jump identity. Diagram T11.4 shows the increasing-jump geometry. A uniformly convex f lies strictly below every nontrivial chord, so only decreasing shocks are allowed.

15. Exact cubic-flux Riemann solution

For $f(u) = u^3 - u$, $a = -1$, seek a shock to a positive state b whose speed equals the left edge of a rarefaction:

$$\frac{f(b) - f(-1)}{b + 1} = b^2 - b = 3b^2 - 1 = f'(b).$$

The nontrivial root is $b = 1/2$ and $\sigma = -1/4$. The entropy solution is

$$u(x,t) = \begin{cases} -1, & x/t < -1/4, \\ \sqrt{(x/t + 1)/3}, & -1/4 < x/t < 2, \\ 1, & x/t \geq 2. \end{cases}$$

On $[-1, 1/2]$, $f(u) - \ell(u) = (u + 1)(u - 1/2)^2 \geq 0$. The shock satisfies the chord criterion and the fan is continuous at $x/t = 2$. Rankine–Hugoniot and the $O(t)$ trace argument prove the weak initial problem. The local scalar contraction theorem proves uniqueness in the bounded entropy class with strong local traces; these Riemann states are not in $L^1(\mathbb{R})$.

16. The perturbed sonic shock is curved

Take $0 < \lambda < 1/4$. Label left characteristics by $\alpha < 0$:

$$a(\alpha) = \begin{cases} -1 + \lambda, & \alpha \le -1, \\ -1 - \lambda\alpha, & -1 < \alpha < 0, \end{cases} \qquad x = \alpha + (3a(\alpha)^2 - 1)t.$$

Their Jacobian in the sloping part is $1 - 6\lambda a t > 0$, so this left region has no characteristic crossings. The initial shock from Problem 15 now encounters increasing left states. Let its position be $s(t)$, its left trace $a(t) < 0$, and its right trace $b(t) = -a(t)/2$. Rankine–Hugoniot and tangency give

$$s' = 3a^2/4 - 1 = f'(b), \qquad s = \alpha + (3a^2 - 1)t, \qquad a = -1 - \lambda\alpha. \tag{11.18}$$

Differentiating the second equation and substituting the first yields

$$\alpha' = -\frac{9a^2}{4(1 - 6\lambda a t)}, \qquad a' = \frac{9\lambda a^2}{4(1 - 6\lambda a t)} > 0.$$

Thus $s'' = (3/2)aa' < 0$: the shock bends to the left relative to its initial line. An explicit parameterization until it reaches the left plateau is, with $r = -a \in [1 - \lambda, 1]$,

$$t(r) = \frac{4}{15\lambda}(r^{-8/3} - r^{-1}), \quad \alpha(r) = \frac{r - 1}{\lambda}, \quad s(r) = \alpha(r) + (3r^2 - 1)t(r).$$

Indeed the linear ODE for $t(a)$ has integrating factor $(-a)^{8/3}$. The plateau is reached at $t_0 = t(1 - \lambda)$; after that $a = -1 + \lambda$ and the shock continues linearly with speed $3(1 - \lambda)^2/4 - 1$, matching its previous position and tangent.

The right-hand fan is no longer entirely centered at the origin. For each emission time $0 \le \theta \le \min(t, t_0)$, the tangent ray

$$x = s(\theta) + s'(\theta)(t - \theta), \qquad u = b(\theta)$$

fills the region from the shock to $x = -t/4$. Since $\partial_\theta x = s''(\theta)(t - \theta) < 0$ before t_0, these rays do not cross in their domain and define the new fan uniquely. For $-t/4 < x < 2t$ retain the original centered fan of Problem 15; for $x > 2t$ retain $u = 1$. On the left use the noncrossing initial characteristics up to the shock. The match at $x = -t/4$ has $u = 1/2$. Each shock jump $a \to -a/2$ obeys $f(u) - \ell(u) = (u - a)(u + a/2)^2 \ge 0$ on that interval, so it is admissible even though the fan derivative can be unbounded on the sonic boundary. The solution is continuous away from the shock, piecewise smooth away from fan edges, and its bounded fluxes permit exhaustion by such regions in the weak identity. Equivalently one

may use the scalar characteristic construction and its continuous fan limit. The initial transition region is bounded with an $O(t)$ perturbation near the origin, yielding the required local trace. For a quantitative early-time check,

$$s(t) = -\tfrac{1}{4}t - \tfrac{27}{16}\lambda t^2 + O(t^3).$$

Keeping the old fan $\sqrt{(x/t+1)/3}$ directly against a curved shock would violate the entropy tangency condition. The emitted rays are an essential part of the perturbation.

17. Smooth scalar contraction in several dimensions

Let $w = u - \hat{u}$ and define the smooth vector field $b = \int_0^1 F'(\hat{u} + \theta w)\, d\theta$. Then $w_t + \operatorname{div}(bw) = 0$. To avoid a discontinuous terminal adjoint, use $\eta_\delta(w) = \sqrt{w^2 + \delta^2} - \delta$. The chain rule gives

$$\partial_t \eta_\delta(w) + \operatorname{div}(b\eta_\delta(w)) = \{\eta_\delta(w) - w\eta'_\delta(w)\} \operatorname{div} b.$$

The expression in braces has absolute value at most δ and is zero where $w = 0$. On a finite time strip with common compact spatial support and smooth bounded derivatives, integrate and use $\delta \int |\operatorname{div} b|$ on that compact set as an error. It tends to zero. Dominated convergence gives $\|w(t)\|_1 = \|w(0)\|_1$, which is stronger than contraction as long as both solutions remain smooth. Uniform compact support on a finite strip follows from bounded speeds and initially compact support in the smooth regime, or may be assumed explicitly in this calculation. After shocks form the smooth renormalization argument no longer applies, and only entropy contraction remains available.

11.6 References

Source attributions

Evans states that T.-P. Liu wrote the initial chapter draft and cites Liu [Lu] as a major reference. The local Riemann construction is attributed to Lax, Comm. Pure Appl. Math. 10 (1957), 537–566. The scalar entropy uniqueness theorem is attributed to S. Kruzhkov, Mat. Sbornik 123 (1970), 228–255, using the source's bibliographic numbering. Further sources listed there include Bressan [Bs], Courant–Friedrichs [C-F], Dafermos [D], Majda [Mj], Xiao–Zhang [X-Z], Zheng [Zh], Logan [Lo], Neu [Ne1], and Smoller [S]. The source acknowledges P. Colella for calculations and W. Han for the eigenvector-dependence proof. Problem 3 is credited to Kato [K]; Problems 4–6 to Holden–Risebro [H-R]; Problems 10–11 to Smoller; Problem 13 to Dafermos. These are attributions read in Evans, not a claim of separate consultation of all the cited works.

Proof dependencies and boundaries

The proofs use inverse/implicit function theorems, smooth ODE theory, spectral algebra, dominated convergence, L^1 translation continuity, and cutoff approximation. Global eigenframes are constructed on $\mathbb{R}^m$; on other state domains the wave geometry is local. The Riemann solver has local amplitude and small-strength uniqueness. The blow-up proof requires a chart covering the entire compact invariant rectangle, a uniform spectral gap, and positive uniform nonlinearity coefficients.

The source refers general viscous-profile sufficiency to Majda–Pego, J. Diff. Eq. 56 (1985), 229–262. Our local proof supplies the scalar reduction but invokes an external center-manifold theorem, which is not proved here. It does not cover distant excursions. The general non-genuinely nonlinear Liu/profile result, referred to Conlon, Adv. Math. 35 (1980), 1–18, remains unproved here. The partially viscous elastic model is proved directly by a scalar ODE.

The viscous limit assumes bounded smooth approximants and a.e. convergence; general-system compactness is not proved. Scalar contraction includes strong initial traces and a local version for Riemann data. General system existence/uniqueness, Glimm/front-tracking theory, compensated compactness, vacuum, multidimensional systems, and nonsmooth system-entropy approximation remain outside the proofs. The 17 solutions and five diagrams are editorial additions. Source comparison, self-check, compilation, and visual QA are distinct from independent mathematical review, which has not been performed.